

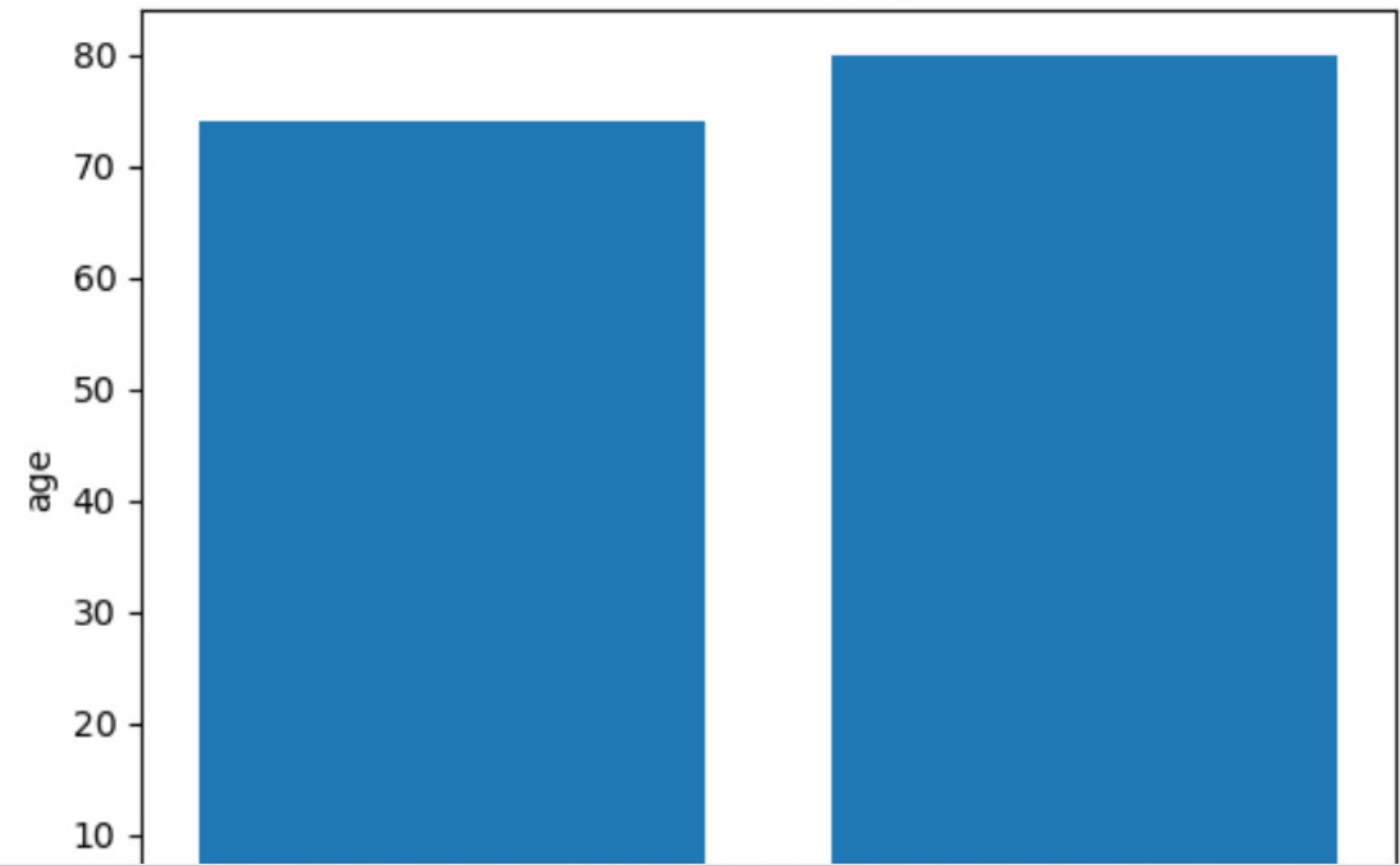
```
[49]: import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("titanic")
print(df.head())
plt.bar(df['survived'],df['age'])
plt.title("survivors by age")
plt.xlabel("survived")
plt.ylabel("age")
plt.show()

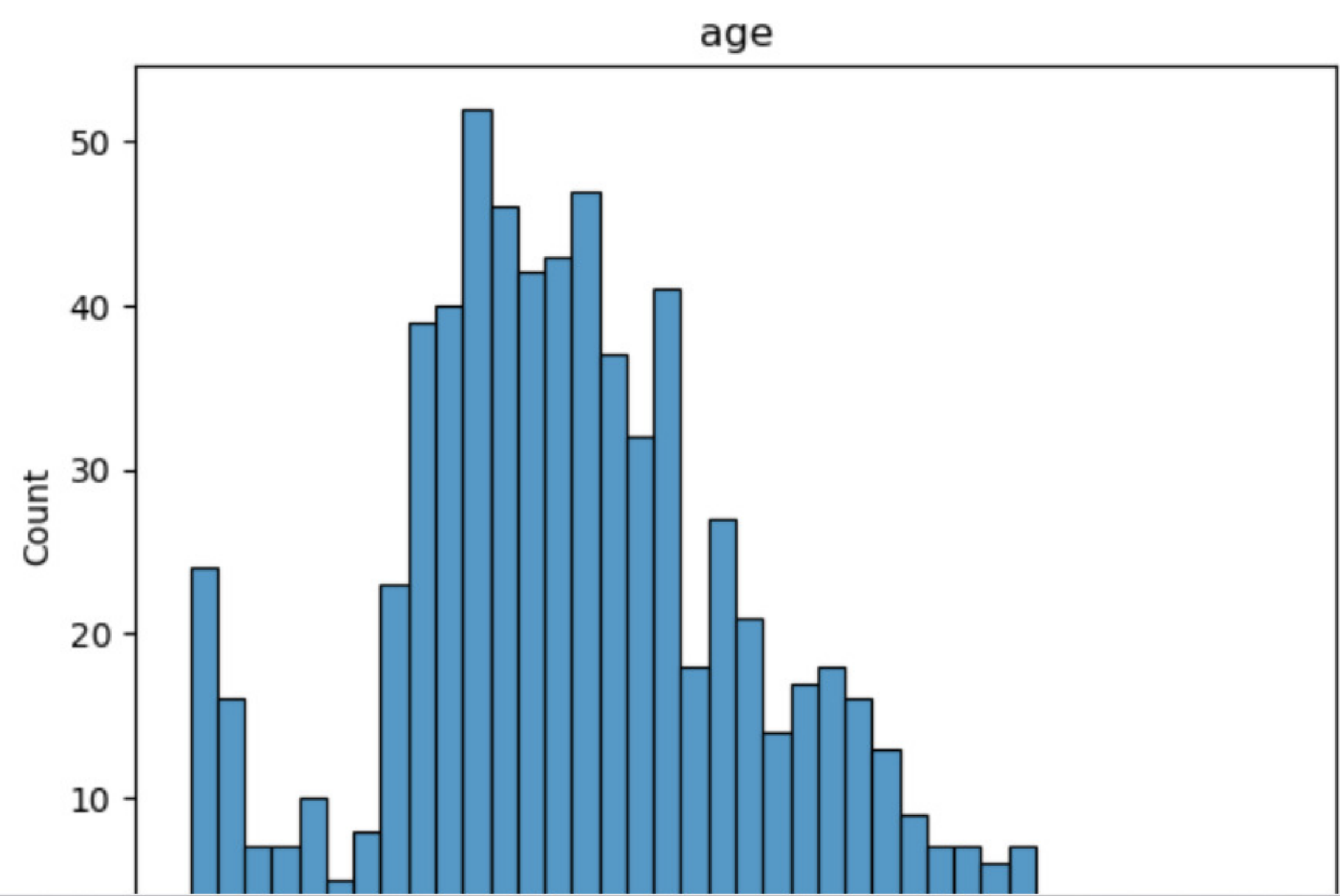
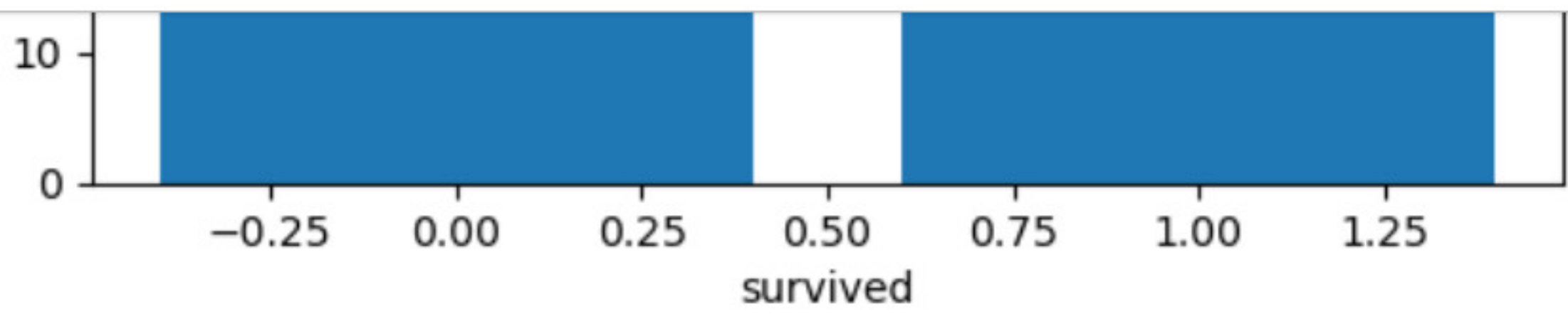
sns.histplot(
    data=df,
    x="age",
    bins=40)
plt.title("age")
plt.show()
```

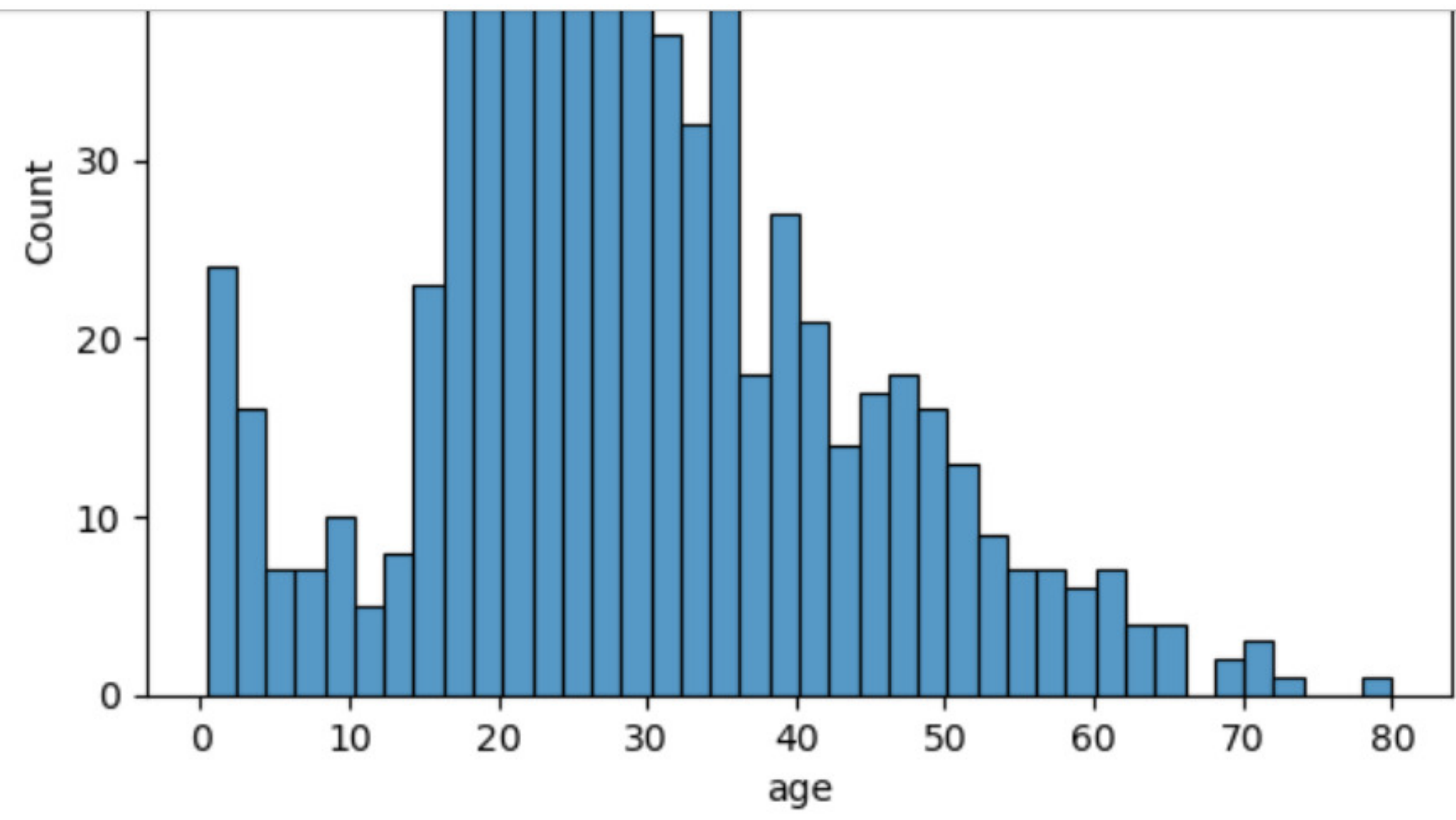
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

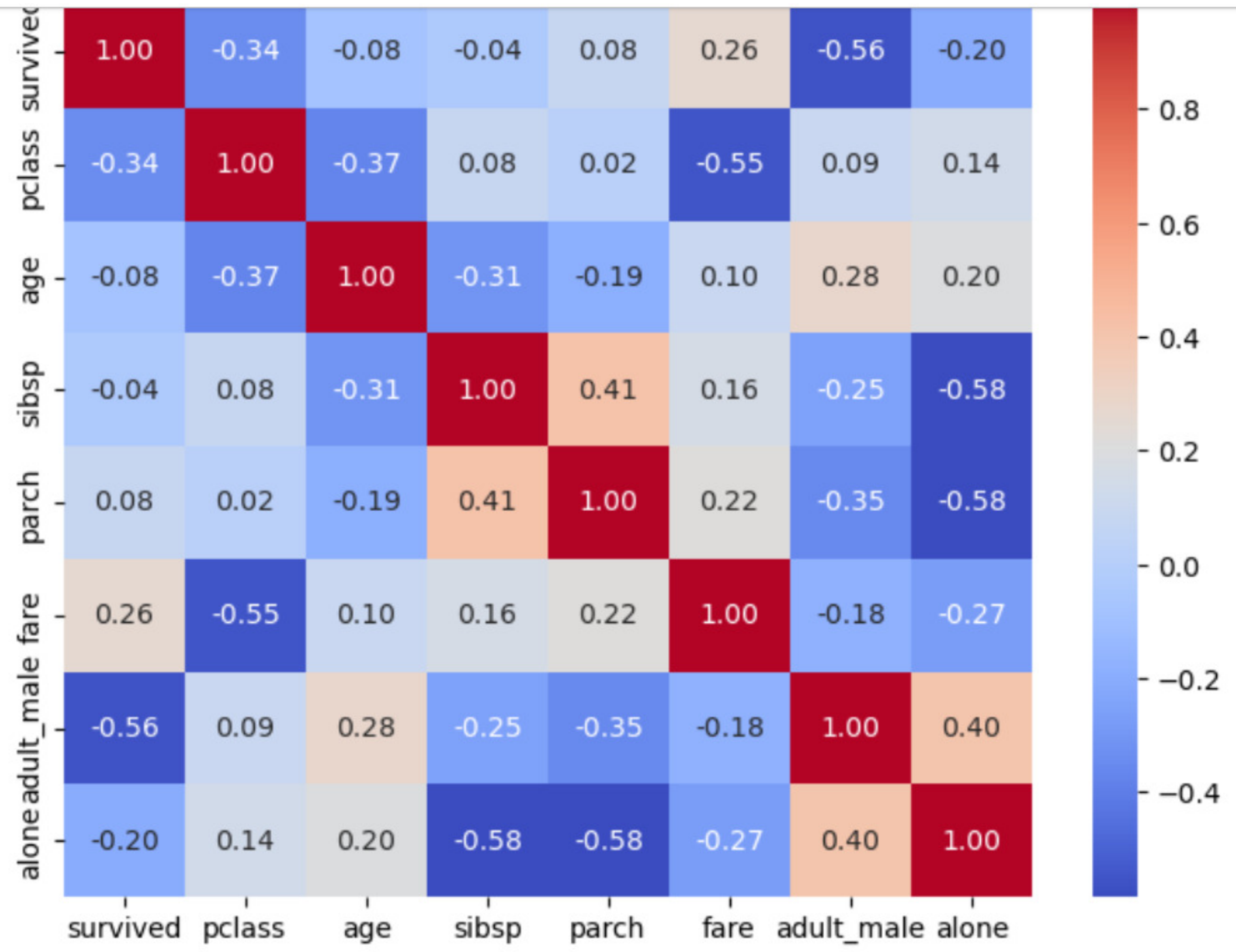
survivors by age







```
[50]: plt.figure(figsize=(8,6))
corr=df.corr(numeric_only=True)
sns.heatmap(corr,
            annot=True,
            cmap='coolwarm',
            fmt='.2f')
plt.show()
```





```
[16]: m=[1,2,24,32,11,14,53]
      mean=sum(m)/len(m)
      print("Mean :",mean)

      Mean : 19.571428571428573
```

```
[24]: #Median
      import statistics
      m=[10,20,60,40,50]
      median=statistics.median(m)
      print("Median :",median)

      #mode
      mo=[1,2,3,2,2,1]
      mode=statistics.mode(mo)
      print("Mode :",mode)

      Median : 40
      Mode   : 2
```

```
[26]: #mean,median,mode using pandas
      import pandas as pd
      df=pd.DataFrame({
          "Marks":[45,50,55,60,90]})
      print(df['Marks'].mean())
      print(df['Marks'].median())
      print(df['Marks'].mode())
```



```
print(dt[ 'Marks' ].mode())
```

```
60.0
55.0
0    45
1    50
2    55
3    60
4    90
Name: Marks, dtype: int64
```

```
[29]: #variance
import numpy as np
data=[10,12,14,16,18]
print(np.var(data))
print(np.mean(data))
print(np.std(data))
```

```
8.0
14.0
2.8284271247461903
```

```
[51]: import numpy as np
data=[10,12,14,16,18,100]
Q1=np.percentile(data,25)
print("Q1 :",Q1)
Q3=np.percentile(data,75)
print("Q3 :",Q3)
IQR=Q3-Q1
```



```
Q3=np.percentile(data,75)
print("Q3 :",Q3)
IQR=Q3-Q1
print("IQR:",IQR)

l=Q1-(1.5*IQR)
u=Q3+(1.5*IQR)
print("Lower limit:",l)
print("Upper limit:",u)

#To find outliers
Outliers=[x for x in data if x<l or x>u]
print("Outliers:",Outliers)
```

```
Q1 : 12.5
Q3 : 17.5
IQR: 5.0
Lower limit: 5.0
Upper limit: 25.0
Outliers: [100]
```

```
[55]: sample_space={1,2,3,4,5,6}
event_A={2,4,6} #Even numbers
print("Sample space (S):",sample_space)
print("Event A:",event_A)

#Probability P(A)
P_A=len(event_A)/len(sample_space)
```



```
#Probability P(A)
P_A=len(event_A)/len(sample_space)
print("\nProbability of event A")
print("P(A)=",P_A)

#conditional probability
#(P|B)
event_B={4,5,6} # Numbers greater than 3
intersection=event_A.intersection(event_B)
P_A_given_B=len(intersection)/len(event_B)
print("\nEvent B:",event_B)
print("A n B:",intersection)
print("P(A|B) =",P_A_given_B)

#Bayes Therom
#example
#Disease Prevalence=1%
#Test Sensitivity=99%
#Positive test probability=5%

P_Disease=0.01
P_Positive_given_Disease=0.99
P_Positive=0.05

P_Positive_given_Disease=(P_Positive_given_Disease*P_Disease)/P_Positive
print("\nBayes Therom Example")
```



```
print("\nBayes Therom Example")
print("P(Disease) =",P_Disease)
print("P(Positive|Disease)=",P_Positive_given_Disease)
print("P(Positive)=",P_Positive)
print("P(Positive|Disease)=",round(P_Positive_given_Disease,4))
print("Percentage=",round(P_Positive_given_Disease*100,2),"%")
```

Sample space (S): {1, 2, 3, 4, 5, 6}
Event A: {2, 4, 6}

Probabitivity of event A
 $P(A) = 0.5$

Event B: {4, 5, 6}
 $A \cap B: \{4, 6\}$
 $P(A|B) = 0.6666666666666666$

Bayes Therom Example
 $P(\text{Disease}) = 0.01$
 $P(\text{Positive}|\text{Disease}) = 0.198$
 $P(\text{Positive}) = 0.05$
 $P(\text{Positive}|\text{Disease}) = 0.198$
Percentage= 19.8 %

[56]: #pearson correlation
import pandas as pd



```
df=pd.DataFrame({
    'Hours':[1,2,3,4,5],
    'Marks':[20,40,50,70,90]
})
corr=df['Hours'].corr(df['Marks'])
print("Pearson Correlation:",corr)
#close to +1
#Strong positive relation
```

Pearson Correlation: 0.9948497511671097

```
[57]: #Spearson Correlation
import pandas as pd
df=pd.DataFrame({
    'Rank_Study':[1,2,3,4,5],
    'Rank_Marks':[2,1,3,5,4]
})
corr=df['Rank_Study'].corr(df['Rank_Marks'],method='spearman')
print("Spearman Correlation:",corr)
```

Spearman Correlation: 0.7999999999999999

```
[58]: #Correlation matrix
import pandas as pd
df=pd.DataFrame({
    'Hours':[1,2,3,4,5],
    'Marks':[20,40,50,70,90],
    'Sleep':[8,7,6,5,4]
```



```
df = pd.DataFrame({
    'Hours': [1,2,3,4,5],
    'Marks': [20,40,50,70,90],
    'Sleep': [8,7,6,5,4]
})
print(df.corr())
```

```
      Hours  Marks  Sleep
Hours  1.00000  0.99485 -1.00000
Marks  0.99485  1.00000 -0.99485
Sleep -1.00000 -0.99485  1.00000
```

```
[68]: #Heatmap
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
df=pd.DataFrame({
    'Hours': [1,2,3,4,5],
    'Marks': [20,40,50,70,90],
    'Sleep': [8,7,6,5,4]
})
sns.heatmap(df.corr(),
            annot=True,
            cmap='magma')
plt.show()

#common cmap values
#plasma
```

```
#common cmap values
#plasma
#coolwarm
#viridis
#YlGnbu
#Blues
#Red
#Greens
#magma
```

